

MetaXtract Selections and Outputs

1 General output

Table 1 summarizes the files produced by MetaXtract. Outputs are grouped into (i) file-level metadata and acquisition methods, (ii) scan-level exports for MS1/MS2, and (iii) interactive visualisations for performance monitoring and comparison.

Table 1: Summary of MetaXtract main outputs and their contents. `<sample>` denotes the RAW base name; `*` denotes an auto-generated timestamp or run-specific suffix.

Output	Typical filename	Content	Scope
File-based details	<code><sample>_info_*.tsv</code>	File-level metadata extracted from the vendor RAW, including instrument identifiers (model/name/serial), software/hardware versions, user ID, acquisition timestamps, scan counts (MS1/MS2 centroid/profile), run time range, mass range, resolution, and other summary statistics. A “Sample Information” block is exported as key-value pairs (e.g., sample id, method paths, injection metadata).	Per RAW file
MS Method	<code><sample>_MS_method.txt</code>	Mass spectrometer acquisition method text, exported directly from the RAW (vendor method definition as plain text). Useful for auditing acquisition settings across runs.	Per RAW file
LC Method	<code><sample>_LC_method.txt</code>	Liquid chromatography method text, exported directly from the RAW (when available). Useful for verifying gradient and pump settings across runs.	Per RAW file
Visualisations of MS1	<code>MS1_report.html</code> and/or <code><sample>_MS1_*.html</code>	Interactive HTML report of MS1 scan-level metrics (e.g., TIC, base peak intensity, total number of peaks, injection time) versus retention time, with optional 2D distributions/histograms depending on selected fields. Intended for quick QC and performance monitoring.	Per RAW file; optional across-sample comparison

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Output	Typical filename	Content	Scope
Visualisations of MS2	MS2_report.html and/or <sample>_MS2_*.html	Interactive HTML report of MS2 scan-level metrics (e.g., TIC, precursor intensity, total number of peaks, injection time, charge state, and other selected header fields) versus retention time, with optional distributions/histograms. Intended for QC, fragmentation performance monitoring, and troubleshooting.	Per RAW file; optional across-sample comparison
Export MS2 extended peak list	<sample>.*parquet	MS2 peaklist export including m/z–intensity arrays per MS2 scan plus extended scan arrays (noises, charges, resolution, baselines). The file is indexed to be mapped back to the meta data file.	Per RAW file
Export MS1 peak list	<sample>.*parquet	MS1 peaklist export including m/z–intensity arrays per MS1 scan.	Per RAW file

2 MS2 Scan Header

When the user selects one or more MS2 parameters, MetaXtract exports an MS2 scan-header table (***scan_header_ms2_*.csv**). Each row corresponds to one MS2 scan (typically centroided MS2), and the selected parameters are exported as columns. Table 2 defines the available MS2 scan-header parameters.

Table 2: MS2 scan-header parameters and their meaning.

Parameter	Meaning
Scan Number	Sequential scan index within the RAW file.
RAW File	RAW base name (source file identifier).
Total Ion Current	Sum of intensities for the scan (scan-level total signal).
Total Number of Peaks	Number of detected peaks in the scan (centroid peaks for centroided data).
Base Peak Mass	m/z of the most intense peak (base peak) in the scan.
Base Peak Intensity	Intensity of the most intense peak (base peak) in the scan.
Retention Time (s)	Scan retention time in seconds.
Elapsed Scan Time (sec)	Instrument-reported scan duration / elapsed time for the scan event (seconds).
Precursor Intensity	Instrument-reported precursor intensity associated with this MS2 scan (searched by the Monoisotopic M/Z in the master scan).
Monoisotopic M/Z	Monoisotopic m/z estimate of the precursor selected for fragmentation.
Charge State	Assigned charge state of the selected precursor (if available).
MS2 Isolation Width	Isolation window width (m/z) used for precursor selection.
Mass Ranges	Acquisition mass range(s) applied to the scan event (scan event definition).
Activation Type	Fragmentation/activation method (e.g., HCD, CID, ETD; instrument-dependent).

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Parameter	Meaning
Collision Energy	Collision energy setting used for activation (unit depends on instrument/method).
HCD Energy	HCD energy (often the normalized collision energy, NCE) used for the scan (if present).
HCD Energy eV	HCD collision energy expressed in electron-volts (if instrument reports it).
Mass Analyzer Type	Mass analyzer used for this scan (e.g., Orbitrap, Ion Trap; instrument-dependent).
Detector Type	Detector used to acquire this scan (instrument-dependent).
Orbitrap Resolution	Orbitrap resolution setting for the scan (if applicable).
AGC	Automatic Gain Control target / setting for ion accumulation (instrument-reported).
Ion Injection Time (ms)	Ion injection / accumulation time used for this scan (milliseconds).
Micro Scan Count	Number of microscans averaged/combined to form this scan.
Number of Channels	Channel count for the scan (instrument- and mode-dependent; e.g., multiplexing).
Frequency	Instrument-reported frequency parameter for the scan (vendor-specific; mode-dependent).
Scan Description	Textual scan event description string (instrument-reported).
Dependency Type	Scan dependency classification (e.g., dependent vs. independent acquisition; vendor-specific).
Master Scan Number	Reference scan number that this scan depends on (e.g., precursor/master scan), if applicable.
API Process Delay	Instrument/API processing delay parameter (vendor-specific timing/overhead metric).
Average Scan by Inst	Instrument-reported running/average scan statistic (vendor-specific; often used for diagnostics).
Access ID	Internal instrument access identifier for the scan/event (vendor-specific).
Multi Inject Info	Multi-injection / repeated injection metadata (only populated when multi-inject is enabled).
Injection t0	Injection time-zero / offset parameter used internally for timing corrections (vendor-specific).
Reagent Ion Injection Time (ms)	Injection time for reagent ion accumulation (only relevant for reagent-ion modes; vendor-specific).
Conversion Parameter I	Internal conversion parameter used by the instrument (vendor-specific).
Conversion Parameter A	Internal conversion coefficient A (vendor-specific).
Conversion Parameter B	Internal conversion coefficient B (vendor-specific).
Conversion Parameter C	Internal conversion coefficient C (vendor-specific).
Conversion Parameter D	Internal conversion coefficient D (vendor-specific).
Conversion Parameter E	Internal conversion coefficient E (vendor-specific).
Temperature Comp. (ppm)	Temperature compensation term in ppm (vendor-specific mass accuracy correction).

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Parameter	Meaning
RF Comp. (ppm)	RF compensation term in ppm (vendor-specific mass accuracy correction).
Space Charge Comp. (ppm)	Space-charge compensation term in ppm (vendor-specific mass accuracy correction).
Resolution Comp. (ppm)	Resolution compensation term in ppm (vendor-specific mass accuracy correction).
Number of LM Found	Number of lock-mass (LM) features detected for correction (if lock mass is used).
LM Correction (ppm)	Lock-mass correction applied (ppm), if lock mass correction is enabled.
RawOvFtT	Raw override/frequency/timing diagnostic value (vendor-specific).
FAIMS Voltage On	Boolean/flag indicating FAIMS was enabled for the scan (if supported).
FAIMS CV	FAIMS compensation voltage (CV) applied to the scan (if supported).

3 MS1 Scan Header

When the user selects MS1 parameters, MetaXtract exports an MS1 scan-header table (`*scan_header_ms1_*.csv`). Each row corresponds to one MS1 scan. Table 3 defines the exported MS1 scan-header parameters.

Table 3: MS1 scan-header parameters and their meaning.

Parameter	Meaning
Scan Number	Sequential scan index within the RAW file.
RAW File	Name of the source RAW file from which the scan originates.
Ion Injection Time (ms)	Ion accumulation (injection) time used for the MS1 scan, reported in milliseconds. Reflects ion population control via AGC.
Total Number of Peaks	Number of detected peaks in the MS1 scan (centroid peak count for centroided data).
Total Ion Current	Sum of all peak intensities within the MS1 scan (total signal intensity).
Retention Time (s)	Chromatographic retention time of the scan, reported in seconds.
Base Peak Intensity	Intensity of the most intense peak (base peak) in the MS1 scan.
Base Peak Mass	m/z value of the most intense peak (base peak) in the MS1 scan.
Multi Inject Info	Instrument-reported metadata related to multi-injection acquisition modes (if enabled).
Multiple Injection	Boolean/flag indicating whether the scan was acquired under multiple-injection conditions.